

Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

AIM:

To model and simulate a simple Transaction-Level Model (TLM) using Initiator and Target modules in SystemC.

Program:

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>

using namespace sc_core;
using namespace tlm;
using namespace std;

// Target
struct Target : sc_module
{
    tlm_utils::simple_target_socket<Target> socket;

    SC_CTOR(Target) : socket("socket")
    {
        socket.register_b_transport(this, &Target::b_transport);
    }

    void b_transport(tlm_generic_payload& trans, sc_time& delay)
    {
        int* data =
            reinterpret_cast<int*>(trans.get_data_ptr());

        cout << "Target received: " << *data << endl;

        delay += sc_time(10, SC_NS);
    }
};

// Initiator
struct Initiator : sc_module
{
    tlm_utils::simple_initiator_socket<Initiator> socket;
```

```

SC_CTOR(Initiator) : socket("socket")
{
    SC_THREAD(thread_process);
}

void thread_process()
{
    tlm_generic_payload trans;

    sc_time delay = SC_ZERO_TIME;

    int data = 555;

    trans.set_data_ptr(
        reinterpret_cast<unsigned char*>(&data)
    );

    trans.set_data_length(sizeof(data));

    cout << "Initiator sending: " << data << endl;

    socket->b_transport(trans, delay);

    wait(delay);
}
};

int sc_main(int argc, char* argv[])
{
    Initiator initiator("initiator");
    Target target("target");

    initiator.socket.bind(target.socket);

    sc_start();

    return 0;
}

```

The screenshot displays the EDA Playground web interface. On the left sidebar, the 'Languages & Libraries' section is active, showing 'Testbench + Design' with 'C++/SystemC' selected. Below it, 'Libraries' includes 'None' and 'SystemC 2.3.3'. The 'Tools & Simulators' section shows 'C++' selected for 'Compile Options' with '-DSC_INCLUDE_FX'. The 'Run Options' section has 'Use run.bash shell script' checked. The main editor area shows a file named 'testbench.cpp' with a single line of code: '1'. The right sidebar shows a file named 'design.cpp' with the following code:

```

12         cout << "Time: " << sc_time_stamp()
13             << " - Clock Triggered" << endl;
14     }
15 }
16
17 SC_CTOR(ClockExample) {
18     SC_THREAD(process);
19     sensitive << clk.pos();
20 }
21 };
22
23 int sc_main(int argc, char* argv[]) {
24     sc_clock clock("clock", 1, SC_NS);
25
26     ClockExample module("module");
27     module.clk(clock);
28
29     sc_start(5, SC_NS);
30
31     return 0;
32 }
33

```

At the bottom, the 'Log' tab is active, showing the compilation and execution output:

```

[2026-08-25 08:18:09 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered

```